Multiobjective Subpixel Mapping With Multiple Shifted Hyperspectral Images

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***Abstract*— Subpixel mapping (SPM) is a useful technique that can interpret the spatial distribution inside mixed pixels and produce a finer-resolution classification map for hyperspectral remote-sensing imagery. However, SPM is essentially an ill-posed problem that requires additional information to produce the unique solution. The limited information of a single image is insufficient to make the mapping problem well posed, whereas the complementary spatial information of multiple shifted images is able to reduce the uncertainty and generate an accurate map. The maximum *a posteriori* model is a feasible way to incorporate auxiliary information for SPM with multiple shifted images, but it introduces a sensitive regularization parameter, which is difficult to preset. Furthermore, the fixed parameter in the iterations influences the incorporation of the multiple images and the spatial prior. In this article, to address these issues, a multiobjective SPM framework for use with multiple shifted hyperspectral images (MOMSM) is proposed. In the proposed algorithm, a multiobjective model consisting of two objective functions, i.e., data fidelity and spatial prior terms, is constructed to transform the SPM into a multiobjective optimization problem, to get rid of the sensitive regularization parameter. To simultaneously optimize the two objective functions, a multiobjective memetic algorithm with a local search operator and an adaptive global replacement strategy is proposed. The multiple images and spatial information can be dynamically fused and the optimal mapping solution with a good balance between the two objectives can be finally obtained. Experiments conducted on both synthetic and real data sets confirm that the proposed method outperforms the other tested SPM algorithms.**

***Index Terms*— Hyperspectral imagery, maximum *a posteriori* (MAP) model, multiobjective optimization, multiple shifted images, remote sensing, subpixel mapping (SPM).**

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I. Introduction

HYPERSPECTRAL remote-sensing imagery has a very high spectral resolution, which is useful in land-cover classification. However, the spatial resolution of hyperspectral imagery is relatively low, resulting in a large number of mixed pixels. Since the spectral curve of a mixed pixel is the combination of spectral signals of different categories, it is inaccurate to simply assign a label to the mixed pixel via hard classification. Therefore, subpixel information extraction is necessary for hyperspectral imagery. The spectral unmixing and subpixel mapping (SPM) techniques can further interpret the class composition and spatial distribution within mixed pixels. Spectral unmixing is a soft classification method designed to determine the components (endmembers) of the mixed pixels and the proportions (fractional abundance) of these components. Compared with hard classification, spectral unmixing can better preserve the spectral information contained in mixed pixels. However, in order to locate the spatial distribution of the endmembers inside a mixed pixel, it is essential to perform SPM after the spectral unmixing.

SPM was introduced by Atkinson *et al.* [1]. It involves dividing a mixed pixel into a predefined number of subpixels and assigning an appropriate class label to each subpixel based on the spatial dependence assumption, i.e., the propensity for nearby locations to influence each other and to possess similar attributes [2]. As a postprocessing step of spectral unmixing, SPM can produce a high-resolution (HR) classification map with more spatial distribution details, based on the low-resolution (LR) abundance maps.

As an effective way to interpret mixed pixels, SPM technology has now been studied for decades. Many SPM algorithms have been proposed based on spatial dependence modeling. For example, Hopfield neural networks (HNNs) [3], [4] and back-propagation (BP) neural networks [5], [6] build an energy function, with the goal of maximizing the spatial dependence. The spatial attraction model utilizes the attraction of adjacent pixels to describe the spatial correlation [7], [8]. Pixel-swapping algorithms make the spatial distribution of subpixels more compact by randomly changing their labels [9]–[11]. The computational intelligence-based methods were designed to obtain the optimal mapping solution. Typical methods include differential evolution (DE) [12], genetic algorithms [13], [14], artificial immune systems [15], multiagent systems

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[16], and multiobjective evolutionary algorithms (MOEAs) [17]. Furthermore, Getis index-based methods [18], Markov random field (MRF)-based methods [19], geostatistical methods [20], [21], and geometric algorithms [22], [23] have also been proposed to incorporate spatial distribution information and alleviate the influence of the unmixing error. What is more, the spatial correlation has been used as a prior term to formulate the SPM as a regularization problem, as in the maximum *a posteriori* (MAP) -based methods [24], [25] and sparse representation-based methods [26].

Although the aforementioned methods have been successfully applied to SPM, they just use the abundance maps derived from a single image. Generally, SPM is an ill-posed problem whose solution is not unique. Since the abundance information of a single image is insufficient to make the SPM problem well posed, the traditional SPM methods mainly rely on the spatial dependence model to constrain the mapping process. However, the spatial dependence hypothesis is not fully consistent with the actual distribution, resulting in ambiguous mapping results. For the purpose of reducing the uncertainty and improving the mapping accuracy, additional auxiliary information derived from multiple images (i.e., multitemporal, multiangle, multiscale, or multisource) can be utilized to solve the ill-posed SPM problem [27]–[31]. The shifted image is one of the auxiliary data for SPM, which refers to the multiple images with a spatial shift at the subpixel scale and it can be obtained by camera movement or satellite orbit translation in the same area. Such images can be obtained by camera movement or satellite orbit translation in the same area. Recently, a number of studies have been conducted on multiple shifted image SPM. For example, Ling *et al.* [28] constructed an HNN model and used multiple shifted images as constraints added to the energy function of the HNN; Wang and Shi [29] utilized multiple shifted images to increase the SPM accuracy, based on fast bilinear and bicubic interpolation; and Peng *et al.* [30] allocated hard class labels to subpixels using the higher resolution multiple shifted images derived from spatial–spectral information. Since the MAP model is a feasible way to integrate the complementary information of the multiple images, Xu *et al.* [31] proposed an SPM framework based on a MAP model with multiple hyperspectral images (MMSSM), which introduces a regularization parameter to fuse the multiple shifted images and the spatial prior information, thus transforming the ill-posed SPM problem into a regularization problem. The optimal mapping solution is then generated via a gradient descent algorithm. Although the experimental results obtained have demonstrated that MMSSM can significantly improve the mapping accuracy, the pivotal regularization parameter needs to be empirically adjusted. To address this issue, Zhong *et al.* [32] presented an adaptive MAP SPM model (AMMSSM), where an L-curve or U-curve method is used for the regularization parameter estimation. However, some problems still exist in these MAP-based SPM methods. First, the regularization parameter of MMSSM is sensitive to the inputs and has a great impact on the mapping accuracy. Even though an adaptive parameter selection strategy was proposed in AMMSSM, it still has problems, such as the low adaptive ability and the sensitivity to the initial values. Second, the regularization parameter in MMSSM and AMMSSM is fixed during the iterations, which affects the fusion of the information of the multiple images and the spatial prior. In addition, both MMSSM and AMMSSM adopt a gradient descent algorithm to optimize the multiple shifted image SPM model, which can get trapped in local optima.

Multiobjective optimization algorithms have been used for resolving regularization-based ill-posed problems [33], such as hyperspectral image classification [34]–[36], sparse unmixing [37], [38], and single-image SPM [33], [39], and they are capable of solving the above-mentioned problems encountered in the MAP-based SPM methods. Therefore, in this article, a multiobjective SPM framework for use with multiple shifted hyperspectral images (MOMSM) is presented. In MOMSM, the multiple shifted image SPM problem is formulated as a multiobjective optimization problem, based on the proposed multiobjective model. Since memetic algorithms with powerful global and local search capabilities provide an effective approach for multiobjective optimization problems [40], [41], a multiobjective memetic algorithm is proposed to optimize the multiobjective mapping model and search for the optimal HR spatial distribution maps.

The main contributions of this article are as follows.

1) A multiobjective model is proposed for SPM with multiple shifted images. The multiobjective mapping model is derived from the MAP-based SPM model and is composed of two objective functions: a data fidelity term and a spatial prior term (Laplacian prior term). The former establishes the correspondence between the multiple LR abundance maps and the HR spatial distribution map. The latter represents the spatial dependence of subpixels. Differing from the MAP-based methods, the use of a sensitive regularization parameter can be avoided in the proposed model.
2) A multiobjective memetic algorithm with MAP-based local search is designed for SPM. In the proposed method, a multiobjective memetic algorithm is designed to optimize the multiobjective SPM model. The candidate HR distribution maps are encoded as solutions. Under the framework of the decomposition-based MOEA (MOEA/D), the DE/rand/1 strategy is utilized for the global search and a MAP-based operator is proposed for the local search, which can quickly improve the quality of the mapping solutions and accelerate the convergence. Because local search may lead to premature convergence of the search process, the mapping solutions are updated by the adaptive global replacement (AGR) strategy, to maintain their diversity. Since the data fidelity and spatial prior terms are simultaneously optimized in the multiobjective memetic algorithm, the complementary information of multiple images, and the spatial correlation, can be intelligently integrated during the optimization.

The rest of this article is organized as follows. In Section II, the basic principles of multiple shifted image SPM are introduced and the motivation for using a multiobjective optimization

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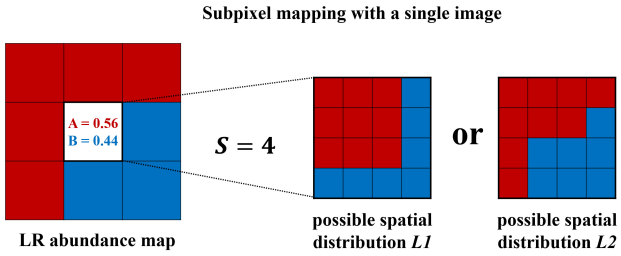


Fig. 1. Single-image SPM produces ambiguous results.

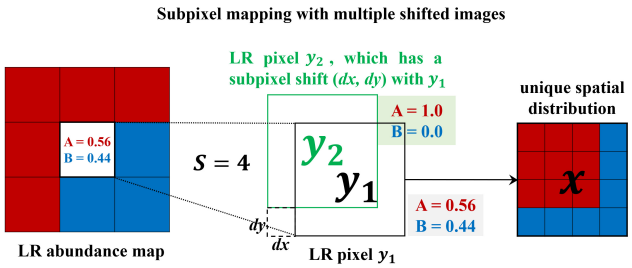


Fig. 2. SPM with multiple shifted images obtains the unique result.

algorithm is explained. Section III describes the details of the multiobjective model and the MOMSM algorithm. The experimental results are presented in Section IV. Further analysis of the control parameters and the optimization algorithm are also provided. Our conclusions are finally drawn in Section V.

II. SPM Background and Motivation

The key issue in SPM is how to produce an HR classification map based on the LR abundance maps. An effective way is to divide each coarse pixel of the LR abundance map into $S \times S$ subpixels and to predict the spatial distribution pattern of these subpixels based on the spatial dependence assumption, where S is the predefined reconstruction scale.

A. SPM With a Single Image

For the single-image SPM, the observation model between the LR abundance map and the HR abundance map of class c is defined as follows:

$$\mathbf{Y}^c = \mathbf{D}\mathbf{X}^c + \mathbf{n}^c 0 \le \mathbf{X}^c \le 1 \tag{1}$$

where the LR abundance map with a size of $[m, n]$ is reshaped to a row vector $\mathbf{Y}^c$ with a size of $[1, mn]$. $\mathbf{D} = [1/S^2, 1/S^2, \ldots, 1/S^2]$ is the down-sampling matrix with a size of $[1, S^2]$. $\mathbf{X}^c$ is the HR abundance matrix with a size of $[S^2, mn]$, which can be reshaped to the HR abundance map with a size of $[M, N], M = S \times m, N = S \times n \cdot \mathbf{n}^c$ is the potential noise or error matrix.

However, as SPM is an ill-posed problem, the limited information of a single image is insufficient to make the mapping problem well posed, resulting in ambiguous results, as shown in Fig. 1. Under the reconstruction scale of 4, the center pixel of the LR abundance map is divided into $4 \times 4 = 16$ subpixels. SPM with a single image will produce two possible spatial distributions, $L1$ and $L2$, which satisfy both the fractional constraint and the spatial dependence assumption.

B. SPM With Multiple Shifted Images

Multiple shifted image SPM can reduce the uncertainty of SPM and improve the mapping accuracy by incorporating shifted data. The basic idea of multiple shifted image SPM is to use the displacement between the multiple LR abundance maps to provide auxiliary spatial information and produce a resolution-enhanced classification map. Fig. 2 provides a simple example, where y_1 is a coarse pixel of the base LR abundance map and y_2 is a pixel of the shifted map that has a subpixel shift (dx, dy) with y_1. According to the fractions provided by y_1 and y_2, the overlapping subpixels are supposed to be class A, and the rest of the subpixels, therefore, belong to class B. Consequently, the distribution of the subpixel block $\mathbf{x}$ can be accurately determined.

Due to the advantages of expansibility and the ease of adding prior information, the MAP model has been successfully utilized for multiple shifted image SPM in the MMSSM method proposed by Xu *et al.* [31]. In MMSSM, the subpixel distributions of the base LR abundance maps are estimated to generate the HR abundance maps. The final classification map can then be obtained according to a class determination strategy. The relationship between the multiple LR abundance maps and the HR abundance maps is defined as

$$\mathbf{Y}_k^c = \mathbf{D}\mathbf{M}_k\mathbf{X}^c + \mathbf{n}^c, \quad k = 1, 2, \ldots, K \tag{2}$$

where $\mathbf{Y}_k^c$ is a column vector with a size of $[mn, 1]$ that represents the kth LR abundance map of class c. $\mathbf{X}^c$ is the HR abundance vector with a size of $[MN, 1]$, $M = m \times S$, $N = n \times S$, and S denotes the reconstruction scale. $\mathbf{M}_k$ is an $[MN, MN]$ warp matrix, which projects $\mathbf{X}^c$ into the coordinates of the kth shifted image. $\mathbf{D}$ is an $[mn, MN]$ down-sampling matrix, and $\mathbf{n}^c$ is the noise matrix. K is the total number of multiple shifted images.

Based on the mapping model (2), the MAP model is used to regularize the ill-posed SPM problem to be well posed by adding some prior information, so that the SPM problem is then converted into a regularization problem. The minimization cost function of MMSSM is, therefore, defined as

$$f\left(\mathbf{X}^c\right) = \sum_{k=1}^{K} \|\mathbf{Y}_k^c - \mathbf{D}\mathbf{M}_k\mathbf{X}^c\|^2 + \lambda U\left(\mathbf{X}^c\right). \tag{3}$$

The cost function can be separated into two parts: $\sum_{k=1}^{K} \|\mathbf{Y}_k^c - \mathbf{D}\mathbf{M}_k\mathbf{X}^c\|^2$ is the data fidelity term, which represents the cumulative reconstruction error of the mapping result and $U(\mathbf{X}^c)$ is the prior term, which represents the spatial dependence information, such as the Laplacian prior model [42] defined in (4). $\mathbf{Q}$ is a 2-D Laplacian operator. A regularization parameter λ is also introduced to combine the data fidelity term with the spatial prior term

$$\text{Laplacian:} U\left(\mathbf{X}^c\right) = \|\mathbf{Q}\mathbf{X}^c\|_2^2. \tag{4}$$

MMSSM adopts the gradient descent algorithm [43] to minimize the cost function (3). The desired HR abundance

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map is solved by the successive approximation iteration in (5), where t denotes the current iteration and α^c is the step size. $\nabla(\cdot)$ is the divergence operator, and $\mathbf{M}_k^T$ is the transpose matrix of $\mathbf{M}_k$

$$\begin{aligned}\mathbf{X}_{t+1}^c &= \mathbf{X}_t^c + \alpha^c \mathbf{r}_t^c \\ \mathbf{r}_t^c &= 2\sum_{k=1}^{K} \mathbf{M}_k^T \mathbf{D}\left(\mathbf{Y}_k^c - \mathbf{D}\mathbf{M}_k\mathbf{X}_t^c\right) - \lambda \nabla U\left(\mathbf{X}_t^c\right). \end{aligned} \quad (5)$$

C. Motivation

From the cost function (3), we can find that λ is a crucial parameter used to balance the effects of the data fidelity term and the spatial prior term. However, λ is sensitive to different data sets and is difficult to preselect. Furthermore, λ is fixed in the iteration of the gradient descent algorithm, which affects the fusion of the two objective functions. An alternative way to solve these problems is to discard the regularization parameter λ and simultaneously optimize the data fidelity term and spatial prior term. Therefore, it is natural to consider the SPM problem as a multiobjective optimization problem.

In this article, a multiobjective optimization algorithm is adopted to solve the multiobjective SPM problem for two reasons. One reason is that the evolutionary multiobjective optimization algorithm has a more powerful optimization ability than the traditional gradient descent algorithm and it is able to simultaneously optimize the multiple objectives. The other reason is that multiobjective optimization methods (e.g., the nondominated sorting genetic algorithm II [44], the MOEA based on decomposition (MOEA/D) [45], and the indicator-based multiobjective optimization algorithm [46]) have been successfully applied to solve the optimization problems in remote-sensing image processing [39], e.g., image clustering [36], sparse reconstruction [47], spectral unmixing [48], [49], and feature selection [50], [51]. Multiobjective optimization for resolving the ill-posed SPM problem is a promising new approach developed only recently. For instance, Ma *et al.* [17] designed a multiobjective framework based on MOEA/D for the single-image SPM (MOSM) and Song *et al.* [52] applied a multiobjective optimization algorithm to solve the sparse SPM problem (MOSSM). However, in MOSM and MOSSM, the multiobjective optimization algorithms are applied to solve the ill-posed problem of the single-image SPM. In the proposed approach, a multiobjective optimization algorithm is adopted to improve the intelligence and accuracy of multiple shifted image SPM.

III. Multiobjective SPM With Multiple Shifted Images

A novel multiple shifted image SPM framework based on multiobjective optimization (MOMSM) is proposed in this article. MOMSM consists of two main steps: 1) the construction of the multiobjective SPM model and 2) the optimization of the proposed model. In the first step, the two objective functions derived from the MAP-based SPM methods are introduced. In the second step, a multiobjective optimization method is introduced, combining MOEA/D with a memetic algorithm. The flowchart of the proposed algorithm (MOMSM) is shown in Fig. 3. The two processes of MOMSM are described as follows.

A. Multiobjective Model for Multiple Shifted Image SPM

MOMSM converts the multiple shifted image SPM problem into a multiobjective optimization problem for the purpose of getting rid of the sensitive regularization parameter. The multiobjective model derived from the cost function (3) consists of two objective functions: 1) the data fidelity term and 2) the spatial prior term, which are defined as shown in the following equations:

$$\begin{aligned} F\left(\mathbf{X}^c\right) &= \min imize\left\{f_1\left(\mathbf{X}^c\right), f_2\left(\mathbf{X}^c\right)\right\} \\ \text{fidelity term}: f_1\left(\mathbf{X}^c\right) &= \sum_{k=1}^{K} \|\mathbf{Y}_k^c - \mathbf{D}\mathbf{M}_k\mathbf{X}^c\|^2 \\ \text{spatial prior term}: f_2\left(\mathbf{X}^c\right) &= U\left(\mathbf{X}^c\right) = \|\mathbf{Q}\mathbf{X}^c\|_2^2. \end{aligned} \quad (6)$$

Due to the contradictory nature of the data fidelity term and the spatial prior term, there is no solution that can make both objectives reach the minimum. Therefore, the optimal solution of the multiobjective model is a set of nondominated solutions with different tradeoffs between the data fidelity term and the spatial prior term. A solution $\mathbf{X}_u$ dominates another solution $\mathbf{X}_v$ if and only if

$$\begin{aligned} &\forall i = 1, 2\ f_i\left(\mathbf{X}_u\right) \le f_i\left(\mathbf{X}_v\right) \\ &\wedge \exists j = 1, 2\ f_j\left(\mathbf{X}_u\right) < f_j\left(\mathbf{X}_v\right). \end{aligned} \quad (7)$$

If a solution $\mathbf{X}_p$ cannot be dominated by the other solutions, then it is a nondominated solution. All the nondominated solutions constitute the Pareto front.

B. Multiobjective Memetic Algorithm for Model Optimization

In the proposed approach, the MOEA (subfitness-based MOEA/D) proposed in MOSM [17] is adopted as the basic multiobjective framework to optimize the proposed SPM model. However, the search of an entire map with complex spatial distribution details is a great challenge. In order to improve the optimization ability, MOEA/D is integrated with a memetic algorithm by adding a local search operator derived from the MAP-based multiple shifted image SPM methods. In addition, the AGR strategy [53] is employed to maintain the diversity of the mapping solutions.

1) Population Initialization: MOEA/D is a population-based optimization algorithm, in which the population is composed of *NP* solutions. For the SPM problem with a reconstruction scale S, its candidate HR abundance map is encoded as a solution $\mathbf{X}^c$ with a length of $M \times N$, as shown in Fig. 4. We can see that the solution $\mathbf{X}^c$ can be divided into $m \times n$ subcomponents $\mathbf{X}^c(j)$ and each subcomponent can be reshaped to a subpixel block of the HR abundance map, which corresponds to a coarse pixel in the LR abundance map.

After inputting the base LR abundance map $\mathbf{Y}^c$, the population $\mathbf{P} = \{\mathbf{X}_1^c, \mathbf{X}_2^c, \ldots, \mathbf{X}_{NP}^c\}$ can be generated by randomly initializing the *NP* candidate solutions around the abundance map $\mathbf{X}_0^c$ produced by the nearest-neighbor interpolation of $\mathbf{Y}^c$

$$\mathbf{X}_i^c = N\left(\mathbf{X}_0^c, 0.01\right), \quad i = 1, 2, \ldots, NP \quad (8)$$

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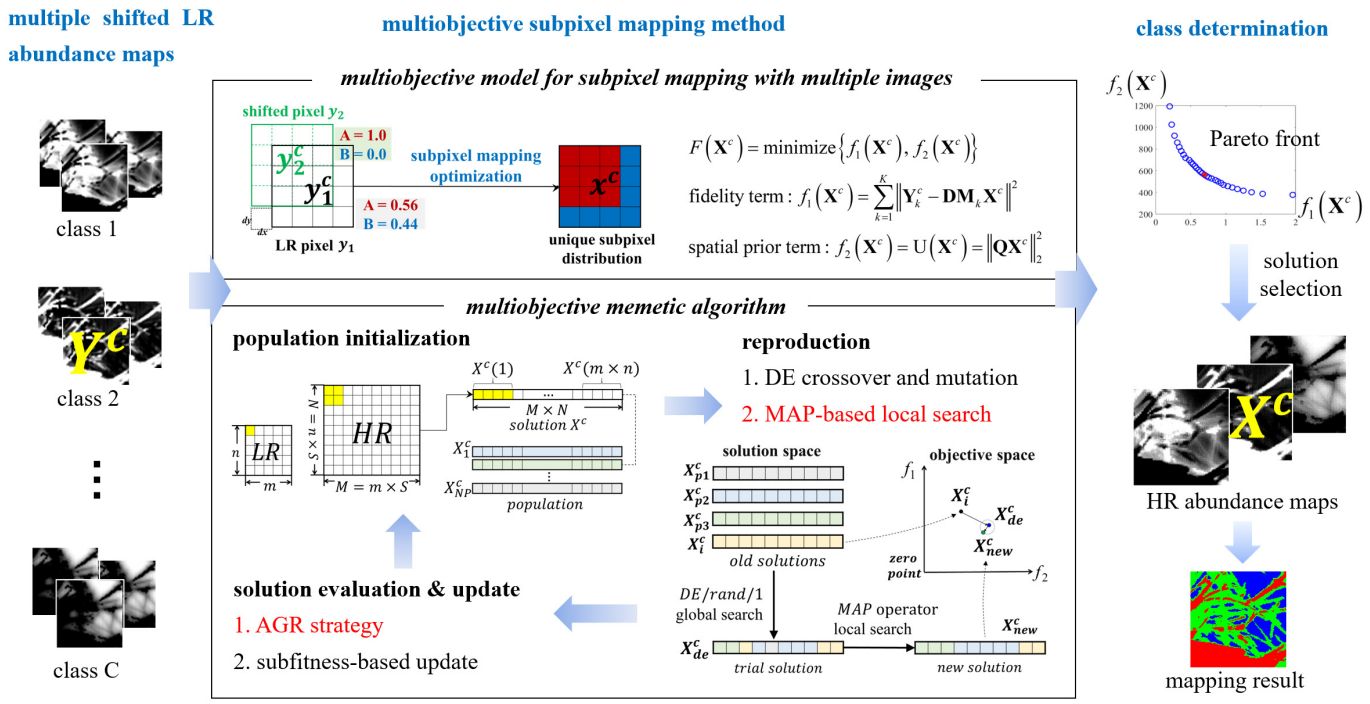


Fig. 3. Flowchart of MOMSM. Input is the multiple shifted LR abundance maps of each class. The multiobjective model for SPM is then constructed. The multiobjective memetic algorithm is employed to minimize the proposed model and search for the optimal Pareto front. The HR abundance maps are selected from the Pareto front and used to determine the final classification map.

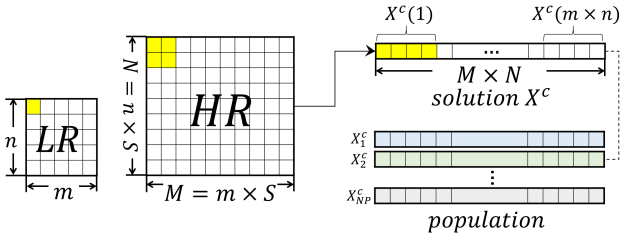


Fig. 4. Candidate HR abundance map is encoded as a solution.

where $\mathbf{X}_i^c$ is the prototype of the ith solution. $N(\mu, \sigma^2)$ is the normal distribution, with an expectation of μ and a standard deviation of σ.

In MOEA/D, the multiobjective optimization problem is decomposed into a number of scalar subproblems, which represent a set of search directions. Each solution is related to a subproblem and is defined by a 2-D weight vector $\lambda^i = [\lambda_1^i, \lambda_2^i]$. These weight vectors $\{\lambda^1, \lambda^2, \ldots, \lambda^{NP}\}$ are uniformly initialized in the objective space and subject to the constraint $\lambda_1^i + \lambda_2^i = 1$. The neighborhood of the weight vector λ^i can then be defined as its T nearest weight vectors $C(\lambda^i) = \{\lambda^{i_1}, \lambda^{i_2}, \ldots, \lambda^{i_T}\}$. The neighborhood set of $\mathbf{X}_i^c$ consists of all the solutions with the weight vectors from the neighborhood of λ^i, i.e., $C(\mathbf{X}_i^c) = \{\mathbf{X}_{i_1}^c, \mathbf{X}_{i_2}^c, \ldots, \mathbf{X}_{i_T}^c\}$.

2) Solution Evaluation: The high-dimensional solution $\mathbf{X}_i^c$ is a separable variable (Fig. 3), so that it can be separately evaluated based on each subcomponent, for which the objective values are used to define the subfitness. MOEA/D uses the Tchebycheff approach to evaluate the solutions. When combined with the multiobjective model (6), the objective value of each subcomponent in solution $\mathbf{X}_i^c$ can be calculated using (9). For the minimization problem, the subcomponent $\mathbf{X}_i^c(j)$ with a lower objective value $F(\mathbf{X}_i^c(j))$ has a higher subfitness, indicating a better quality

$$\begin{aligned} F\left(\mathbf{X}_i^c(j)\right) &= g^{te}\left(\mathbf{X}_i^c(j) \,|\, \lambda^q, \mathbf{z}^*(j)\right) \\ &= \max_{d=1,2}\left\{\lambda_d^q |f_d\left(\mathbf{X}_i^c(j)\right) - \mathbf{z}_d^*(j)|\right\} \\ f_1\left(\mathbf{X}_i^c(j)\right) &= \sum_{k=1}^{K} \|\mathbf{Y}_{k,j}^c - \mathbf{D}\mathbf{M}_k\mathbf{X}_i^c(j)\|^2 \\ f_2\left(\mathbf{X}_i^c(j)\right) &= \|\mathbf{Q}\mathbf{X}_i^c(j)\|_2^2 \end{aligned} \tag{9}$$

where $\mathbf{X}_i^c(j)$ is the jth subcomponent in solution $\mathbf{X}_i^c$. λ^q is the weight vector of the qth subproblem, and $\mathbf{z}^*(j)$ is the reference point of the jth subcomponent, which represents the minimum objective value of the jth subcomponent found so far, i.e., $\mathbf{z}_d^*(j) = \min_{i=1}^{NP}\{f_d(\mathbf{X}_i^c(j))\}, d = 1, 2$. $\mathbf{Y}_{k,j}^c$ is the jth pixel of the kth shifted LR abundance map.

Based on the multiobjective model (6), an objective space with two dimensions is constructed, for which the vertical axis represents the data fidelity term and the horizontal axis refers to the spatial prior term. After evaluation, the solution $\mathbf{X}_i^c$ can be mapped to a point in the objective space, for which the vertical coordinate value is the sum of $f_1(\mathbf{X}_i^c(j))$ and the horizontal coordinate value is the sum of $f_2(\mathbf{X}_i^c(j))$. For the minimization of the mapping model (6), the multiobjective optimization algorithm aims to produce new solutions that are closer to the zero point in the objective space.

3) Reproduction With MAP-Based Local Search: Under the framework of MOEA/D, a memetic algorithm, which is a

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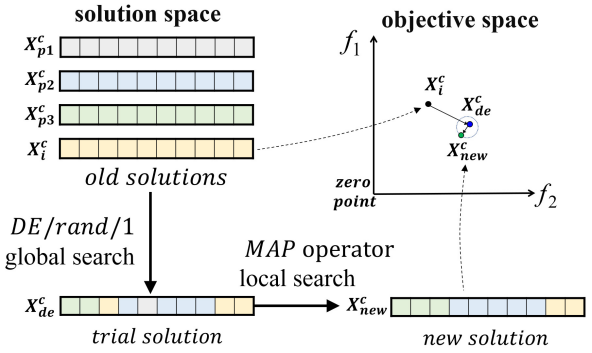


Fig. 5. Reproduction of a new solution with global search and local search.

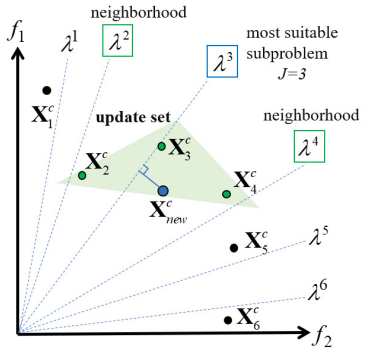


Fig. 6. New solution is used to update the closest solutions.

union of population-based search algorithms, hybridized with local improvements, is adopted to reproduce the new mapping solutions. To be specific, the DE algorithm is utilized for the global search and the MAP-based operator is proposed for the local search. A toy example of the reproduction process is displayed in Fig. 5, which shows the generation of a new solution by global search and local search.

Since the DE algorithm [54] has a powerful global search capability, its DE/rand/1 strategy, formulated as shown in (10), is selected to perform a rough search and locate the promising regions around the solution $\mathbf{X}_i^c$, generating a trial mapping solution $\mathbf{X}_{de}^c$

$$\mathbf{X}_{mu}^c = \mathbf{X}_{p1}^c + F \times \left(\mathbf{X}_{p2}^c - \mathbf{X}_{p3}^c\right)$$
$$\mathbf{X}_{de,j}^c = \begin{cases} \mathbf{X}_{mu,j}^c, & \text{if (rand} \le \text{CR or } j = j_{\text{rand}}) \\ \mathbf{X}_{i,j}^c, & \text{otherwise} \end{cases} \tag{10}$$

where $\mathbf{X}_{p1}^c$, $\mathbf{X}_{p2}^c$, and $\mathbf{X}_{p3}^c$ are three distinct solutions randomly selected from the neighborhood set of $\mathbf{X}_i^c$. The scale factor (F) and crossover rate (CR) are the two control parameters of the DE algorithm, which are set to constants of 0.8 and 0.2, respectively. j_{rand} is a random integer between 1 and NP.

Although the global search can quickly locate the promising mapping solutions in the solution space, it is difficult for it to fine-tune the mapping solutions, due to its limited local search capability. Therefore, for the local search, a MAP-based operator is proposed to refine the trial map $\mathbf{X}_{de}^c$. This operator is inspired by evolutionary algorithm-based fuzzy clustering [55], [56], which demonstrates that the quality of the solutions can be further improved by the fuzzy update after the global search. Although the iteration of the MAP-based SPM method may fall into local optima in a deviating search direction, it can provide an accurate search in a promising search direction. Therefore, the MAP-based local search operator derived from the iterative formulation (5) can be expected to accelerate the generation of high-quality solutions that are closer to the zero point in the objective space. The MAP-based local search operator proposed to update the trial solution $\mathbf{X}_{de}^c$ and produce a new solution $\mathbf{X}_{\text{new}}^c$, which is defined as follows:

$$\mathbf{X}_{\text{new}}^c = \mathbf{X}_{de}^c + \alpha^c \mathbf{r}^c$$
$$\mathbf{r}^c = 2\sum_{k=1}^{K} \mathbf{M}_k^T \mathbf{D}\left(\mathbf{Y}_k^c - \mathbf{D}\mathbf{M}_k \mathbf{X}_{de}^c\right) \tag{11}$$

where the step size α^c is usually set to 0.6. However, the MAP-based operator is a little different from the iterative formulation (5). Since the local search does not have to strictly correct the solution, the item with the regularization parameter can be removed.

4) Population Update With AGR: After performing the reproduction, a new solution X_{new}^c can be obtained, whose subfitness is evaluated via the objective function (9). It can then be used for the population update. In the proposed approach, we employ the AGR strategy [53] to adaptively adjust the update set, i.e., we use the new solution $\mathbf{X}_{\text{new}}^c$ to update its closest solutions, as they are likely to have similar optimal results. The AGR strategy is aimed at maintaining the diversity of the solutions and preventing premature convergence of the population.

First, the new solution $\mathbf{X}_{\text{new}}^c$ is assigned to its most suitable subproblem λ^J using (12). The update scope is adaptively determined using (13), where T is the size of the neighborhood set. itr and $\text{itr}_{\max}$ are the current iteration and maximum iterations, respectively.

$$J = \arg\min_{j=1}^{NP} \left\{ g^{te}\left(\mathbf{X}_{\text{new}}^c \middle| \lambda^j, \mathbf{z}^*\right)\right\} \tag{12}$$

$$\text{Tr} = \lceil \frac{T}{1 + \exp\left(-20 \times (\text{itr}/\text{itr}_{\max} - 0.25)\right)} \rceil. \tag{13}$$

The update set $B(\mathbf{X}_{\text{new}}^c)$ of the solution $\mathbf{X}_{\text{new}}^c$ can then be obtained by selecting the first Tr solutions from the neighborhood set $C(\mathbf{X}_J^c)$ of the Jth subproblem, i.e., $B(\mathbf{X}_{\text{new}}^c) = C(\mathbf{X}_J^c) = \{\mathbf{X}_{J_i}^c\}, i = 1, 2, \ldots, \text{Tr}$. A simple example of the AGR strategy with $\text{Tr} = 3$ is provided in Fig. 6.

In order to update the high-dimensional solution $\mathbf{X}_{J_i}^c$, a subfitness-based update strategy [17] is employed to compare the corresponding subcomponents of $\mathbf{X}_{\text{new}}^c$ and $\mathbf{X}_{J_i}^c$, and to retain those subcomponents with a higher subfitness, i.e., a lower objective function value,

$$\mathbf{X}_{J_i}^c(p)' = \begin{cases} \mathbf{X}_{\text{new}}^c(p), & \text{if } F\left(\mathbf{X}_{\text{new}}^c(p)\right) < F\left(\mathbf{X}_{J_i}^c(p)\right) \\ \mathbf{X}_{J_i}^c(p), & \text{otherwise} \end{cases} \tag{14}$$

where $\mathbf{X}_{\text{new}}^c(p)$ and $\mathbf{X}_{J_i}^c(p)$ are the pth subcomponents of $\mathbf{X}_{\text{new}}^c$ and $\mathbf{X}_{J_i}^c$, and $p = 1, 2, \ldots, mn$. In this way, a child solution $\mathbf{X}_{J_i}^{c\prime}$ with higher quality is produced and preserved in the new population.

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For each solution in the old population $\mathbf{P} = \{\mathbf{X}_1^c, \mathbf{X}_2^c, \ldots, \mathbf{X}_{NP}^c\}$, we repeat the reproduction and population update process until a generation is completed and a new population with NP better solutions is formed. The evolutionary process is repeated until the number of iterations reaches the maximum, i.e., $\text{itr}_{\max} = 20$.

5) Solution Selection: The nondominated solutions of the final population constitute the optimal Pareto front in the objective space, which provides multiple alternative mapping solutions, i.e., HR abundance maps that represent different tradeoffs between the two objective functions. In other words, the multiobjective memetic algorithm has searched for a diverse set of mapping solutions that can ensure the integrity of the objective space. The Pareto front directly provides the visual abundance maps for the user, so that the user can easily make a decision based on these meaningful and intuitive solutions, according to their needs. The selected solution $\mathbf{X}^c$ is the HR abundance map of class c. Therefore, compared with those methods that preset the regularization parameter without prior knowledge, MOMSM can reduce the difficulty of user intervention and the manual selection can make full use of the tradeoff solutions. Furthermore, to enhance the automatic processing capability of MOMSM, the final HR abundance map for class c can also be determined by the angle-based knee point selection strategy [47]. Specifically, we use B-splines to fit the Pareto front and then use an angle-based method [57] to select the knee point (the point with the largest angle) from the Pareto front as the final solution $\mathbf{X}^c$. MOMSM with the knee point selection strategy is denoted as MOMSM$*$.

6) Class Determination: For all the classes $c = 1, 2, \ldots, C$, the multiobjective optimization steps are performed so that we can acquire a set of HR abundance maps $\mathbf{X} = [\mathbf{X}^1, \mathbf{X}^2, \ldots, \mathbf{X}^C]$. Finally, these HR maps are integrated to generate the HR classification map $\mathbf{CM}$ via the winner-takes-all class determination strategy [25], defined as shown in (15), where $i = 1, 2, \ldots, MN$.

$$\mathbf{CM}_i = \arg\max \left\{ \mathbf{X}_i^c \,|c = 1, 2, \ldots, C \right\}. \tag{15}$$

C. Computational Complexity Analysis

We suppose that the LR hyperspectral image with a size of $[m, n]$ contains C land-cover classes and the reconstruction scale of the SPM is S. The computational complexity of the MOMSM algorithm can then be split into four parts: 1) the initialization of the population; 2) the reproduction of new solutions; 3) the calculation of the objective function; and 4) the update process and class determination. Since the population consists of NP solutions and the length of a solution is $L = m \times n \times S^2$, the computational complexity for part 1) is $O(NP \times L)$. The reproduction step is composed of the DE/rand/1 strategy and the MAP local search, whose computational complexity is $O(NP \times L \times G)$, where G denotes the maximum iterations. For each solution, its objective function value of each subcomponent is calculated pixel-by-pixel, so the computational complexity for part 3) is $O(NP \times m \times n \times G)$. As for part 4), the complexity of the population update and class determination is $O(NP \times L \times G)$ and $O(L)$, respectively. Since only one class is optimized at a time in MOMSM, the whole computational complexity should be multiplied by C. Therefore, taking all the aforementioned parts into account, the complexity of the proposed method is $O(NP \times L \times G \times C)$.

IV. Experiments and Analysis

The performance of MOMSM was investigated with both synthetic and real data sets. Two types of synthetic data sets were tested to better evaluate the mapping accuracy, one of which was accurate synthetic data without potential errors (i.e., the Washington DC data set and the Indian Pines data set) and the other type contained unmixing error (i.e., the Xiaqiao data set). Moreover, a real hyperspectral data set polluted by coregistration and spectral unmixing errors (i.e., the Nuance data set) was also used to verify the reliability of the proposed method in a practical situation.

The proposed method was compared with both classical and state-of-the-art methods, including five single-image SPM algorithms and three multiple shifted image SPM methods. The hard classification maps are also provided for comparison.

Hard Classification: The class label with the largest abundance value is assigned to the mixed pixel.

SASM (2006) [7]: A single-image SPM method based on the spatial attraction model.

ADESM (2012) [12]: A single-image SPM method based on the spatial attraction model and the adaptive DE algorithm.

AMCDSM (2015) [25]: A single-image SPM method based on the MAP model, which uses the data fidelity term (1) and the spatial prior term (4) to regularize the SPM problem.

MOSM (2018) [17]: A single-image SPM method based on the multiobjective optimization algorithm.

MOSSM (2019) [52]: A single-image SPM method based on sparse representation and multiobjective optimization.

SMMI (2010) [28]: A multiple shifted image SPM method based on the HNN model, whose energy function takes the multiple images into account.

MMSSM (2013) [31]: A multiple shifted image SPM method based on the MAP model, which combines the data fidelity term (2) with the spatial prior term (4) by a manually defined regularization parameter.

AMMSSM (2014) [32]: An extension of MMSSM with an adaptive regularization parameter.

MOMSM$*$ and MOMSM are the proposed algorithms. The knee point strategy differentiates MOMSM$*$ from MOMSM. A sensitivity analysis of the control parameters of MOMSM is also presented to demonstrate its stability. In addition, analysis of the optimization algorithm is provided to validate the effectiveness of the critical strategies and the optimal solutions.

The accuracy of the mapping results is assessed via two classical indices, i.e., the overall accuracy (OA) and the Kappa coefficient. Furthermore, for the accurate synthetic data sets, the adjusted OA (OA$*$) and the adjusted Kappa (Kappa$*$) [12] calculated only for mixed pixels are also reported to provide an indication of the ability of the different SPM algorithms to handle mixed pixels.

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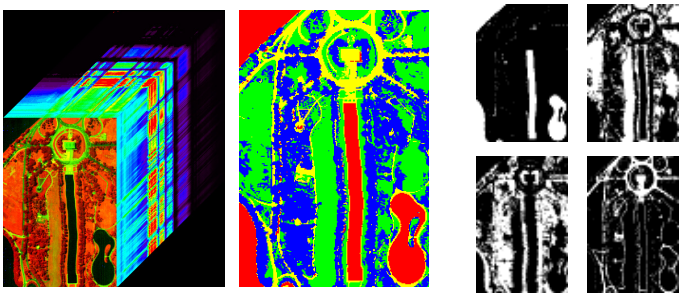


Fig. 7. Synthetic Washington DC data set. (a) Original hyperspectral image (300 × 200 × 192). (b) Reference map. (c) Base LR abundance maps (75 × 50) obtained by down-sampling.

A. Washington DC Data Set

For the synthetic Washington DC data set, the original hyperspectral image [Fig. 7(a)] acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) is made up of 192 bands, 300 lines, and 200 columns, with four categories of water, tree, road, and bare soil. The highly accurate classification map for the original hyperspectral image, produced by the support vector machine (SVM) algorithm, was utilized as the reference map [Fig. 7(b)]. The base LR abundance maps [Fig. 7(c)] were obtained by down-sampling the reference map via a mean filter. Four sets of LR abundance maps with shifts of (–0.5, 0), (0.5, 0), (0, –0.5), and (0, 0.5) relative to the base LR maps were used as the auxiliary data. Given the reconstruction factor of 4, the size of the base LR abundance maps was [75, 50].

The mapping results for the Washington DC data set are shown in Fig. 8. Thanks to the accurate LR abundance maps, the classification maps of the different methods are very close to the ground truth. Comparing the single-image SPM methods (i.e., SASM, ADESM, AMCDSM, MOSM, and MOSSM) with the multiple shifted image SPM methods (i.e., SMMI, MMSSM, and AMMSSM), it can be found that the limited information of the single image leads to mapping uncertainty, resulting in many false predictions, while the Supplementary Information of the multiple shifted images contributes to the accurate reconstruction of the spatial distribution details. Although the results of MMSSM, AMMSSM, and MOMSM are visually similar, a small difference exists in the black box area, which shows that MOMSM achieves a better balance between detail preservation and spatial homogeneity. Although MOMSM∗ and MOMSM use different selection strategies, their mapping results are almost the same.

The quantitative results for the Washington DC data set reported in Table I are consistent with the visual results, i.e., the performance of the multiple shifted image mapping methods is significantly better than that of the single-image mapping methods. MOSM, MOSSM, and MOMSM are based on a multiobjective optimization method, but the OA of MOMSM is 7.5% higher than that of MOSM and 6.5% higher than that of MOSSM. Moreover, due to the powerful optimization ability of the enhanced MOEA, MOMSM outperforms the other methods using multiple shifted images in all four indicators.

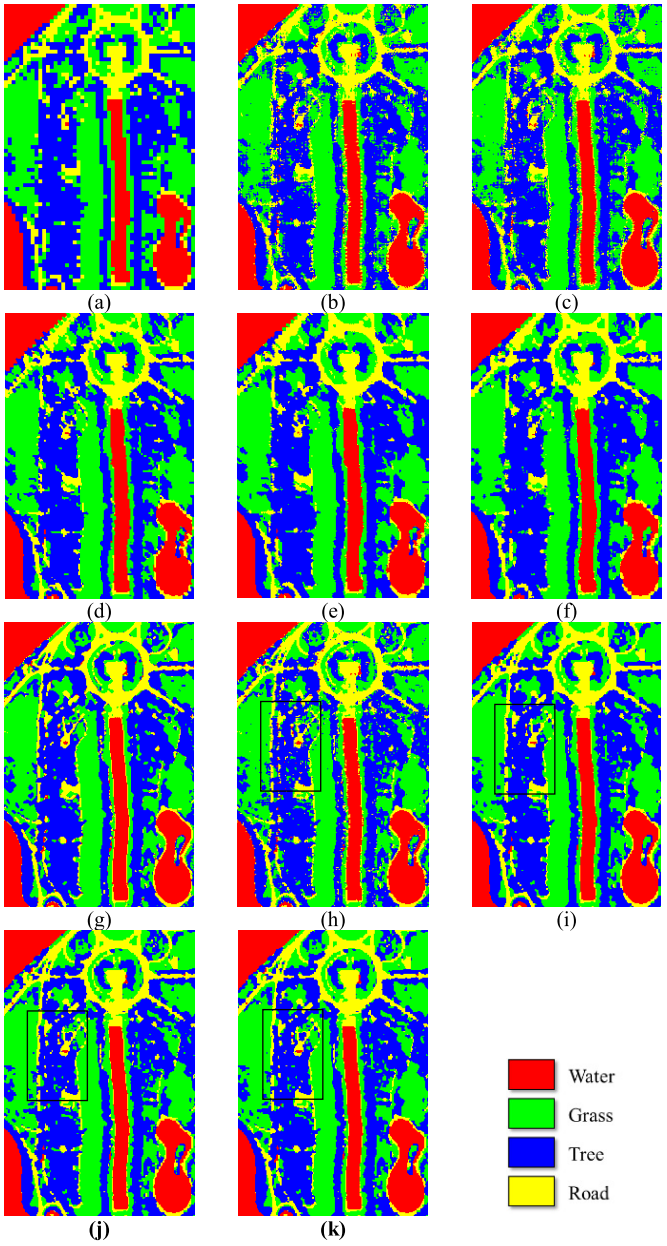


Fig. 8. Mapping results for the Washington DC data set. (a) Hard classification. (b) SASM, 2006. (c) ADESM, 2012. (d) AMCDSM, 2015. (e) MOSM, 2018. (f) MOSSM, 2019. (g) SMMI, 2010. (h) MMSSM, 2013. (i) AMMSSM, 2014. (j) MOMSM∗. (k) MOMSM.

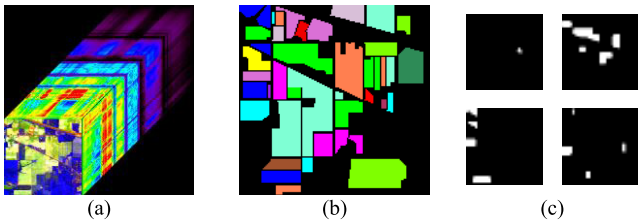


Fig. 9. Synthetic Indian Pines data set. (a) Original hyperspectral image (136 × 136 × 200). (b) Reference map. (c) Four of the 17 base LR abundance maps (34 × 34) obtained by down-sampling.

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TABLE I
MAPPING ACCURACIES FOR THE WASHINGTON DC DATA SET

	OA(%)	Kappa	OA*(%)	Kappa*
Hard	82.6767	0.7524	70.4043	0.5577
SASM	85.8500	0.7994	75.8257	0.6427
ADESM	86.0550	0.8023	76.1760	0.6479
AMCDSM	84.7650	0.7827	74.2483	0.6160
MOSM	84.7267	0.7818	74.6014	0.6200
MOSSM	85.7600	0.7968	75.6720	0.6366
SMMI	91.2500	0.8760	85.1300	0.7810
MMSSM	91.4050	0.8781	85.3161	0.7830
AMMSSM	91.0483	0.8727	84.7124	0.7727
MOMSM*	**92.0667**	**0.8873**	**86.4465**	**0.7988**
MOMSM	**92.2233**	**0.8895**	**86.7141**	**0.8027**

B. Indian Pines Data Set

As with the Washington DC data set, for the synthetic Indian Pines data set, we used the multiple LR abundance maps down-sampled from the HR reference map as input. The original HR hyperspectral image [Fig. 9(b)] acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) is composed of 136 × 136 × 200 pixels and 17 classes. Based on the original image, the reference map [Fig. 9(b)] classified by the SVM algorithm could be obtained. The base LR abundance maps with a size of [34, 34] were obtained by down-sampling the reference map under the reconstruction scale of 4, and four of them are shown in Fig. 9(c). The offsets of the shifted images were (–0.5, 0), (0.5, 0), (0, –0.5), and (0, 0.5). From the reference map, we can see that the variety of land covers with tight but unconnected spatial distributions brings difficulty to the SPM.

Fig. 10 displays the full mapping results for the Indian Pines data set. From the visual results, it can be seen that AMCDSM and MOSM are overdependent on the spatial prior term, resulting in expansion of the borders, which indicates the limitation of single-image SPM. The result of SMMI is too smooth at the corners of these regular shapes. MMSSM has to adjust the regularization parameter to obtain a satisfactory map. Although the adaptive strategy enables AMMSSM to calculate the regularization parameter, it cannot ensure an optimal result, due to the fixed parameter. MOMSM does not only get rid of the sensitive regularization parameter but it also dynamically fuses the data fidelity term and the spatial prior term. Therefore, AMMSSM does not perform as well as MOMSM in dealing with the tight but unconnected objects, as shown in the white box area.

The mapping accuracies for the Indian Pines data set are listed in Table II. From the mapping accuracies, we can see that AMCDSM, MOSM, and MOSSM obtain relatively poor OAs of 91.25%, 90.77%, and 91.55%, respectively, due to their dependence on the spatial priors. From the perspective of OA∗ and Kappa∗, the advantage of SMMI, MMSSM, AMMSSM, and MOMSM is more obvious, which reflects the importance of the multiple shifted images in handling mixed pixels. Since MOMSM can automatically integrate the auxiliary information and spatial prior information during the optimization process, it achieves the highest OA of 98.03%.

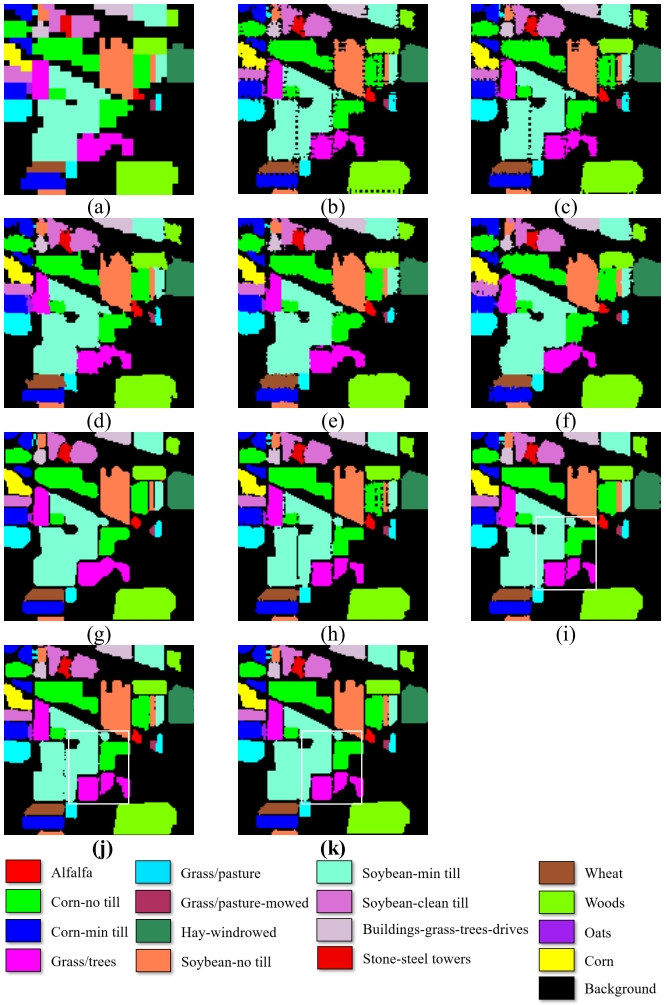


Fig. 10. Mapping results for the Indian Pines data set. (a) Hard classification. (b) SASM, 2006. (c) ADESM, 2012. (d) AMCDSM, 2015. (e) MOSM, 2018. (f) MOSSM, 2019. (g) SMMI, 2010. (h) MMSSM, 2013. (i) AMMSSM, 2014. (j) MOMSM∗. (k) MOMSM.

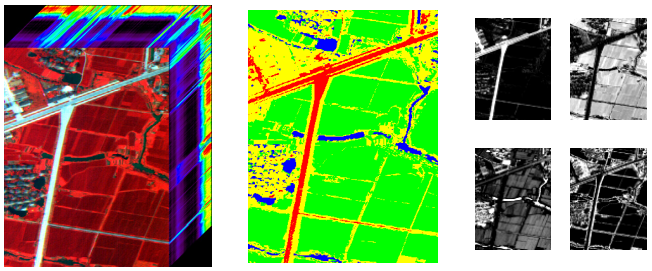
TABLE II
MAPPING ACCURACIES FOR THE INDIAN PINES DATA SET

	OA(%)	Kappa	OA*(%)	Kappa*
Hard	87.8568	0.8436	68.7361	0.6352
SASM	93.8852	0.9185	84.2567	0.8036
ADESM	95.2314	0.9364	94.3486	0.9275
AMCDSM	91.2576	0.9126	78.3964	0.7394
MOSM	90.7710	0.8783	78.9254	0.7435
MOSSM	91.5495	0.8884	78.2851	0.7342
SMMI	96.5700	0.9550	91.6400	0.8950
MMSSM	97.3994	0.9653	93.3046	0.9165
AMMSSM	97.0210	0.9605	92.4415	0.9066
MOMSM*	**97.7076**	**0.9695**	**94.1676**	**0.9276**
MOMSM	**98.0320**	**0.9738**	**94.9749**	**0.9376**

C. Xiaqiao Data Set

Differing from the above synthetic data sets, the LR abundance maps for the Xiaqiao data set were produced by spectral unmixing, instead of down-sampling. Therefore, the

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(a) (b) (c)

Fig. 11. Synthetic Xiaqiao data set. (a) Original hyperspectral image (400 × 300 × 78). (b) Reference map. (c) TBase LR abundance maps obtained by FCLS unmixing (100 × 75).

TABLE III
MAPPING ACCURACIES FOR THE XIAQIAO DATA SET

	OA(%)	Kappa	OA*(%)	Kappa*
Hard	76.1183	0.5753	76.0334	0.5665
SASM	67.1150	0.4739	66.9047	0.4640
ADESM	67.7128	0.4816	67.0889	0.4675
AMCDSM	79.1892	0.6290	79.1470	0.6218
MOSM	79.6800	0.6378	79.6446	0.6309
MOSSM	79.8942	0.6401	79.8618	0.6333
SMMI	74.6642	0.5834	74.5327	0.5758
MMSSM	82.5350	0.6888	82.5394	0.6836
AMMSSM	82.3917	0.6862	82.3940	0.6809
MOMSM*	**83.1558**	**0.6998**	**83.1689**	**0.6949**
MOMSM	**83.2333**	**0.7013**	**83.2474**	**0.6965**

experiment with the Xiaqiao data set was subject to unmixing error. The original hyperspectral image [Fig. 11(a)] of the Xiaqiao data set was collected with an airborne push-broom hyperspectral imager (PHI) from the city of Changzhou in China. There are 400 × 300 × 78 pixels in the original hyperspectral image. The SVM classification map of the hyperspectral image was utilized as the reference map [Fig. 11(b)], which contains four land-cover classes: road, corn/vegetable, water, and bare soil. The multiple hyperspectral images with shifts of (–0.5, 0), (0.5, 0), (0, –0.5), and (0, 0.5) were used as the Supplementary Data. Under the reconstruction scale of 4, the synthetic LR hyperspectral images were obtained through down-sampling the multiple hyperspectral images. The multiple shifted abundance maps [Fig. 11(c)] for the Xiaqiao data set were then produced by conducting spectral unmixing with the fully constrained least squares (FCLS) algorithm [58] on the LR hyperspectral images. As shown in Fig. 11(c), there are obvious unmixing errors in the abundance maps of the Xiaqiao data set. Many pure pixels covered by corn/vegetable are misidentified as water. The mapping of the Xiaqiao data set is a challenging task not only because of the unmixing error but also because of the narrow objects (roads and bare soil) contained in this data set.

The mapping results for the Xiaqiao data set are displayed in Fig. 12. Since SASM, ADESM, and SMMI are sensitive to data error, their mapping results are affected by the unmixing error, i.e., some subpixels are misclassified as water. Although the SPM methods based on a single image (AMCDSM, MOSM, and MOSSM) can reduce the influence of the unmixing error by the spatial prior term (4), they fail to reconstruct the narrow objects, due to the limited abundance information. In contrast, the narrow objects in the results of MMSSM, AMMSSM, MOMSM∗, and MOMSM appear relatively natural, because these methods not only consider the spatial dependence information but also fuse the complimentary information provided by the multiple images. Furthermore, we can see from the detail regions that MOMSM∗ and MOMSM perform better than the other methods in the reconstruction of the narrow objects, which proves that the multiobjective memetic algorithm achieves a good balance between the data fidelity term and the spatial prior term.

The mapping accuracies for the Xiaqiao data set are presented in Table III. Due to the large amounts of unmixing errors in the abundance maps, the OA∗ and Kappa∗ statistics contain great uncertainty, so their values are unreliable. For this inaccurate data set, we can refer to the OA and Kappa of the entire classification map. Influenced by the accumulation of the unmixing error and the SPM error, the accuracy of some methods (SASM, ADESM, and SMMI) even drops below the accuracy of the hard classification map. However, from the detail maps, we can see that some distribution details have been recovered in the maps obtained by these SPM methods. In addition, the accuracy advantage of multiple-image SPM over single-image SPM is not as obvious as in the previous two experiments. MMSSM, AMMSSM, MOMSM∗, and MOMSM all obtain close accuracies. However, MOMSM∗ and MOMSM are free of the sensitive regularization parameter. Since the four algorithms adopt the same data fidelity term and spatial prior term, the improved accuracy of MOMSM∗ and MOMSM proves that the multiobjective algorithm performs better in dealing with the SPM problem.

D. Nuance Data Set

Since multiple shifted images generated by satellite orbit translations in the same area are difficult to obtain, we constructed a real data set by the use of camera movements on the ground. This experiment with a real data set was affected by coregistration and spectral unmixing errors, because the real LR hyperspectral image sequences and the HR digital color image were collected from the same area and FCLS unmixing was used to produce the multiple LR abundance maps. To be specific, four LR hyperspectral images with a size of [50 × 50 × 46] were collected by a Nuance near-infrared (NIR) imaging spectrometer. The base image is shown in Fig. 13(c). The shifts of the other three LR images were calculated by the method proposed in [59] and were (0.31, 0.13), (0.06, 0.15), and (0.20, 0.56). There were three classes in the LR images. Four groups of LR abundance maps produced via FCLS were utilized as the input. The base images are shown in Fig. 13(d). An HR color image of 150 × 150 pixels was acquired at the same time using a digital camera, as shown in Fig. 13(a). The HR reference map could then be obtained by SVM, as displayed in Fig. 13(b). The reconstruction scale was set to 3.

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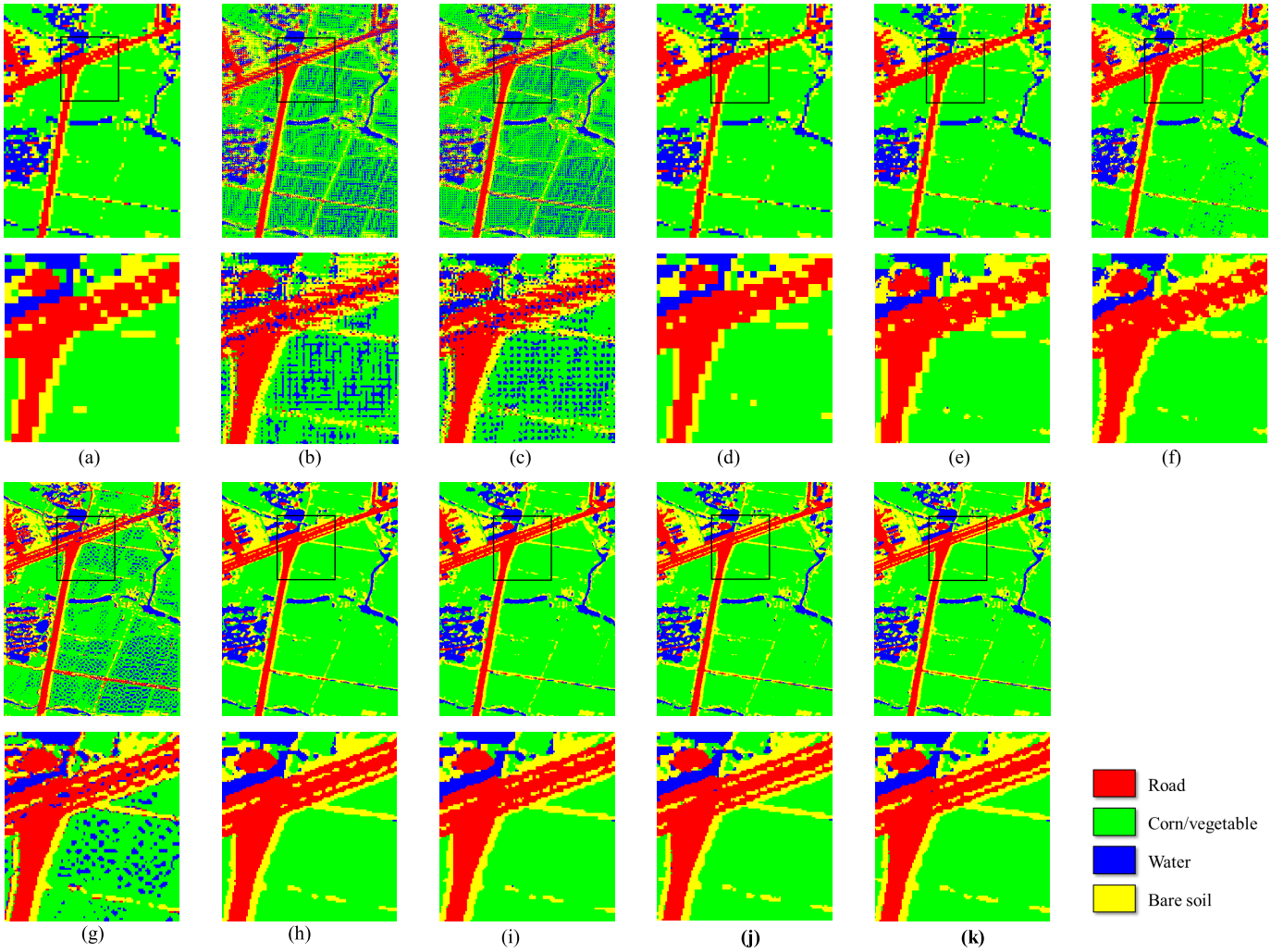


Fig. 12. Mapping results and detail regions for the Xiaqiao data set. (a) Hard classification. (b) SASM, 2006. (c) ADESM, 2012. (d) AMCDSM, 2015. (e) MOSM, 2018. (f) MOSSM, 2019. (g) SMMI, 2010. (h) MMSSM, 2013. (i) AMMSSM, 2014. (j) MOMSM∗. (k) MOMSM.

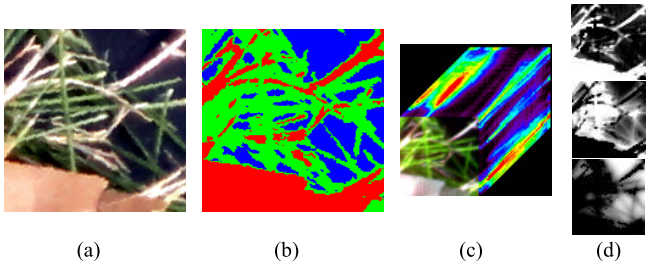


Fig. 13. Nuance data set. (a) Original HR color image (150 × 150 × 3). (b) Reference map. (c) Original Nuance hyperspectral image (50 × 50 × 46). (d) Base LR abundance maps (50 × 50) obtained by FCLS unmixing.

TABLE IV
MAPPING ACCURACIES FOR THE NUANCE DATA SET

	OA(%)	Kappa	OA*(%)	Kappa*
Hard	76.0578	0.6339	66.0428	0.4254
SASM	71.6000	0.5659	58.5933	0.3235
ADESM	71.1867	0.5602	57.9026	0.3233
AMCDSM,	76.2178	0.6371	66.3102	0.4359
MOSM	76.2889	0.6376	66.4216	0.4333
MOSSM	76.9111	0.6464	67.4688	0.4518
SMMI	63.9911	0.4790	53.1194	0.1874
MMSSM	77.1600	0.6530	67.8402	0.4626
AMMSSM	76.3067	0.6395	66.6889	0.4432
MOMSM*	**77.0444**	**0.6508**	**67.6768**	**0.4597**
MOMSM	**77.2267**	**0.6534**	**67.9664**	**0.4641**

A visual comparison of the mapping results for the Nuance data set is provided in Fig. 14 and the accuracies are listed in Table IV. SASM and ADESM preserve many details of the fresh grass class. However, these methods are subject to potential errors because of their strict adherence to the fractional abundance constraint, resulting in many wrongly mapped subpixels. In particular, SMMI performs poorly in this real-data experiment because it is sensitive to the image registration errors. The MAP-based methods (MMSSM and AMMSSM) are more robust to the potential errors but produce oversmoothed results. In contrast, MOMSM generates smoother results in homogeneous regions, while preserving more details around the borders, so that it achieves the highest accuracy. Affected by the large potential errors of the

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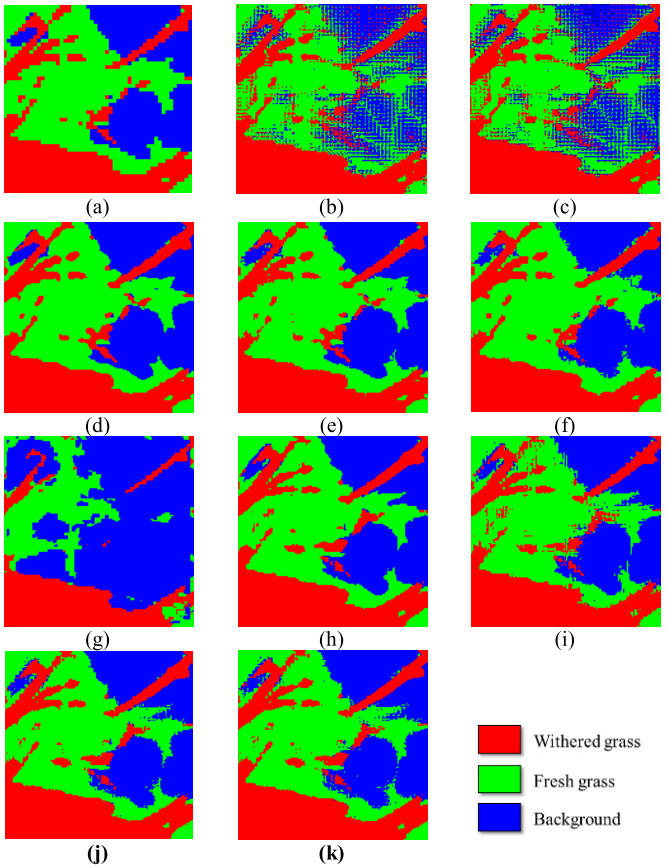


Fig. 14. Mapping results for the Nuance data set. (a) Hard classification. (b) SASM, 2006. (c) ADESM, 2012. (d) AMCDSM, 2015. (e) MOSM, 2018. (f) MOSSM, 2019. (g) SMMI, 2010. (h) MMSSM, 2013. (i) AMMSSM, 2014. (j) MOMSM∗. (k) MOMSM.

abundance maps, all these methods fail to predict the distribution details of the fresh grass class. As a result, the accuracies of some of the SPM methods are even worse than the hard classification.

E. Parameter Sensitivity Analysis

In the proposed multiobjective optimization algorithm, there are several important parameters used to control the DE/rand/1 strategy, including the scale factor F, the CR, and the population size NP. Unified parameters were used in the previous experiments, i.e., $F = 0.8$, CR $= 0.5$, and $NP = 30$. To demonstrate the robustness of MOMSM with regard to different parameter settings within the appropriate range, a further sensitivity analysis was conducted for F, CR, and NP using all the above data sets.

Two parameters were set to be constants when analyzing the other one. For example, the scale factor F was varied from 0.2 to 1.0 at a rate of 0.2, with CR $= 0.5$ and $NP = 30$. The OAs of MOMSM for the four data sets are shown in Fig. 15(a)–(d). It can be found that MOMSM achieves a higher accuracy when $F > 0.6$. However, the difference in the OA is within 0.2%. The impact of different CR values in the range of [0.1, 0.5] on the mapping accuracy is shown in Fig. 15(e)–(h), where the variation in OA is less than 0.2%. We recommend setting CR around 0.4, which can ensure the information exchange in the population. MOMSM with

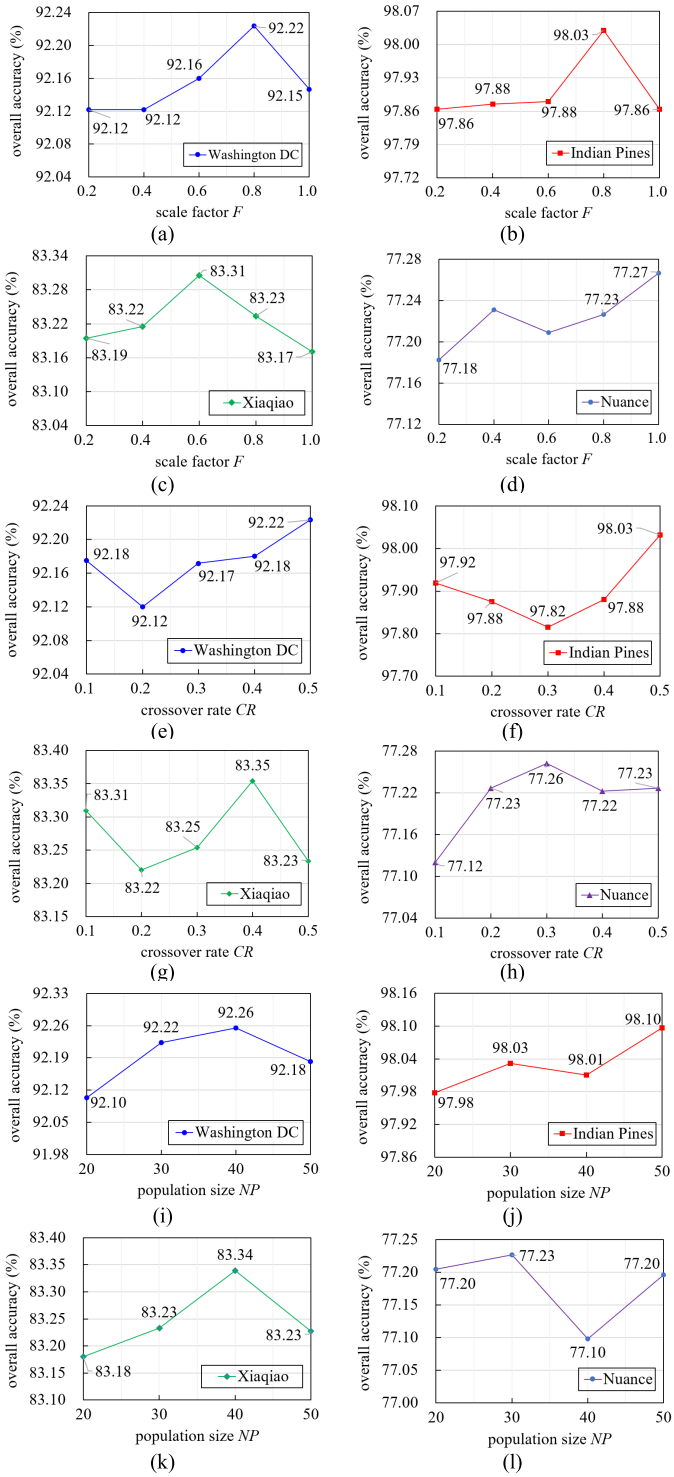


Fig. 15. Parameter sensitivity analysis for MOMSM. (a)–(d) Sensitivity analysis of the scale factor F. (e)–(h) Sensitivity analysis of the CR. (i)–(l) Sensitivity analysis of the population size NP.

a population size ranging from 20 to 50 was also tested, as shown in Fig. 15(i)–(l). Generally, the performance of MOMSM is relatively stable and it is not sensitive to these parameters. One reason for this is that the parameters were set within the appropriate range. Another reason is that the multiobjective optimization algorithm can produce optimal solutions with good tradeoffs to reduce the dependence of the SPM method on parameters.

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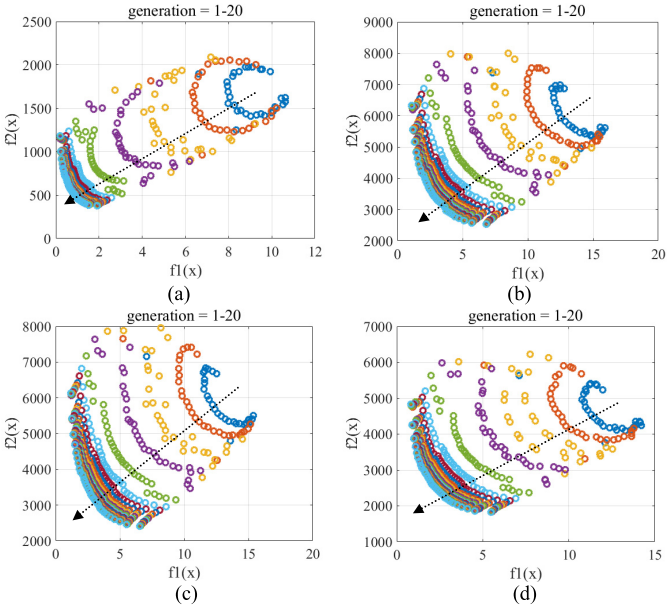


Fig. 16. Population convergence in the objective space during the optimization of each HR abundance map of the Washington DC data set. The dotted line indicates the direction of convergence. (a) Water. (b) Grass. (c) Tree. (d) Road.

F. Optimization Analysis

To validate the effectiveness of the multiobjective memetic algorithm of MOMSM, a further analysis of its optimization procedure was conducted on the Washington DC data set. The populations in the objective space during the optimization of each HR abundance map are displayed in Fig. 16. The populations (composed of 30 mapping solutions) for generation $=$ 1 are marked as the blue circles farthest from the zero point, while the populations for the final generation $=$ 20 are marked as the azure circles closest to the zero point. Since the multiobjective memetic algorithm aims to generate the optimal solutions (HR abundance maps) that minimize the multiobjective mapping model (6), the population can be expected to evolve toward the zero point in the objective space. As shown in Fig. 16, the multiobjective memetic algorithm successfully evolves the populations, which gradually approach the zero point after each generation. In addition, the population tends to converge after 15 generations and these balanced mapping solutions constitute the Pareto front. After 20 generations, the optimization of the population has reached a stable state. The final Pareto front consisting of all the mapping solutions is uniformly distributed in the objective space, which represents a good balance between the data fidelity term and spatial prior term.

There are two important strategies in the proposed multiobjective memetic algorithm: the MAP-based local search operator for solution reproduction and the AGR strategy for population update. These strategies are aimed at enhancing the optimization ability of the proposed method. As mentioned in Section III-B, the MAP-based local search operator is used to accelerate the generation of high-quality solutions that are closer to the zero point in the objective space. The AGR strategy is used to maintain the diversity of the solutions and prevent premature convergence of the population. To demonstrate the effectiveness of these strategies, we took the Washington DC data set as an example. The optimization process of two variants without MAP local search (MOMSM-noMAP) and without the AGR strategy (MOMSM-noAGR) was compared with that of MOMSM. The results are displayed in Fig. 17, where the populations of MOMSM, MOMSM-noMAP, and MOMSM-noAGR for generation $=$ 1, 5, and 20 are shown. The three algorithms were all initialized with the same population. We can see that the populations of MOMSM for the four classes are much closer to the zero point than those of MOMSM-noMAP, which proves that the MAP strategy can quickly improve the quality of the mapping solutions. It is also worth noting that the populations of MOMSM-noAGR are much better than those of MOMSM after one generation. The reason for this is that the AGR strategy limits the update of the mapping solutions at the beginning of the optimization to prevent premature convergence of the population, so MOMSM-noAGR converges faster in the early stage. However, the populations of MOMSM gradually approach (generation $=$ 5) and finally surpass (generation $=$ 20) those of MOMSM-noAGR, which proves the effectiveness of the AGR strategy.

G. Computation Time

As analyzed in Section III-C, the computational complexity of the proposed MOMSM and MOMSM$*$ can be considered as being relatively high. Since MOMSM and MOMSM$*$ share the same optimization procedure, they only differ in the final selection step. The computation time of MOMSM is affected by the manual selection. Therefore, the computation time of MOMSM$*$ with the automated selection strategy is provided here for comparison. In fact, the computational cost is a common problem in evolutionary algorithms and it exponentially increases with the dimension of the solution (image size). To further illustrate this problem, the runtimes of the SPM methods are listed in Table V. All the experiments were conducted using a personal computer with a 2.3-GHz CPU and 8 GB of RAM. We can see that the classical SASM is very efficient, due to its simple principle. AMCDSM and MMSSM are also quite efficient. It should be noted that AMMSSM is as time consuming as the proposed MOMSM$*$ algorithm, because AMMSSM adopts an adaptive strategy to adjust the regularization parameter. As for the evolutionary algorithm-based SPM methods (ADESM, MOSM, MOSSM, and MOMSM$*$), their computational cost is relatively high, because the optimization of these methods is based on the population, i.e., these methods simultaneously search for multiple mapping solutions. In addition, their optimization process becomes particularly time consuming when solving high-dimensional mapping solutions, i.e., their computation time is positively related to the image size. Furthermore, the computation time of MOSM, MOSSM, and MOMSM$*$ is also positively related to the number of component classes. Since MOMSM$*$ adopts the MAP local search strategy and the AGR strategy, its computation time is greater than that of MOSM and MOSSM. Among the multiple shifted image SPM methods, SMMI is computationally efficient, relatively speaking, but it is affected by the poor quality of the abundance

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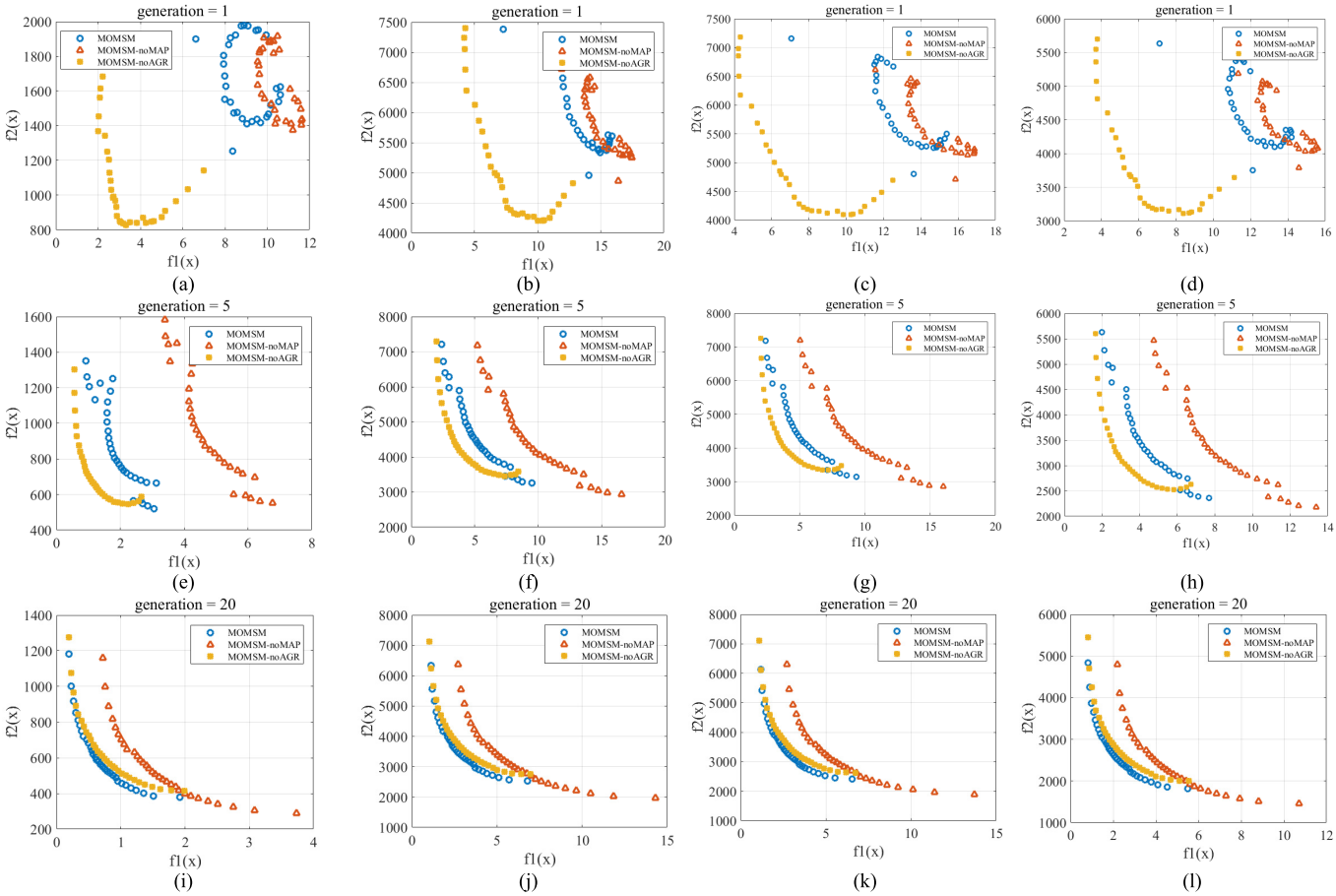


Fig. 17. Comparison of the optimization process of MOMSM, MOMSM without MAP local search, and MOMSM without the AGR strategy. The SPM of the Washington DC data set is taken as an example. Populations of generation = 1, 5, and 20 are shown. (a) Water, generation = 1. (b) Glass, generation = 1. (c) Tree, generation = 1. (d) Road, generation = 1. (e) Water, generation = 5. (f) Glass, generation = 5. (g) Tree, generation = 5. (h) Road, generation = 5. (i) Water, generation = 20. (j) Glass, generation = 20. (k) Tree, generation = 20. (l) Road, generation = 20.

TABLE V

COMPUTATION TIME (S) OF THE SPM METHODS FOR PROCESSING DIFFERENT DATA SETS

	SASM	ADESM	AMCDSM	MOSM	MOSSM	SMMI	MMSSM	AMMSSM	**MOMSM***
Washington DC	0.82	**5223.20**	61.34	835.99	775.54	226.95	26.98	3225.34	3004.09
Indian Pines	1.29	937.15	41.37	1098.30	804.49	276.55	17.57	**2576.09**	**2553.10**
Xiaqiao	1.28	**9464.44**	170.26	1165.51	1663.80	411.10	73.58	4696.22	7278.80
Nuance	0.47	**1883.76**	16.89	355.26	319.16	**1740.46**	4.68	1274.53	833.11

maps, and its network takes a long time to converge when dealing with the Nuance data set. In conclusion, the computation time is a shortcoming of all the evolutionary algorithm-based SPM methods. Under the current hardware conditions, these methods can be accelerated by optimizing each solution of the population in parallel. However, we believe that with the ongoing development of both hardware (computing speed) and software (computing strategy), the evolutionary SPM methods have the potential to be used in practical applications.

V. CONCLUSION

In this article, a multiobjective SPM method with multiple shifted images has been proposed to solve the problem of the sensitive regularization parameter in the traditional methods. In MOMSM, the mapping problem is formulated as a multiobjective optimization problem and the optimal classification map capable of balancing the data fidelity term and the spatial prior term can be obtained by the multiobjective memetic algorithm. The experimental results obtained with four data sets showed that MOMSM produces better mapping results with a higher accuracy. The sensitivity test also indicated that the performance of MOMSM is relatively stable when the parameters are constrained within a reasonable range.

However, the proposed method does have some limitations. 1) As evolutionary optimization involves an iterative process, it is generally slow and struggles to cope with high-dimensional image data. The efficiency of the evolutionary multiobjective optimization algorithm could, thus, be improved by taking advantage of its property of implicit parallelism. 2) The SPM is affected by the quality of the abundance maps and the unmixing errors are passed on to the mapping result. In our future work, we will attempt to improve the computational efficiency and reduce the effect of unmixing errors on the SPM.

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